

Technical; Project Report: Native Wazun SIEM on ARM64
Date: 12-20-2025

Executive Summary

The challenge: Deploying a central Wazuh SIEM stack on NVIDIA DGX Spark (ARM64) hardware.

The Conflict: Official Docker Images and standard automated scripts often failed on ARM due to QEMU emulation segmentation faults and hardware-specific GPG key expirations.

The Solution: A native AARCH64 Installation combined with advanced package manager surgery to bypass broken post-install scripts.

Architecture Overview

Wazuh Agent: Collects endpoint data (Logs, FM, Rootkit detection).

Wazuh Manager (Server): The “brain” that decodes logs and triggers alerts via the Analysis Engine.

Wazuh Indexer: A highly scalable engine that stores alerts in a searchable database (Port 9200).

Wazuh Dashboard: The web UI used for visualization and agent management (Port 443).

Deep-Dive Troubleshooting

Issue A: The GPG Key (Wall)

Symptom: “apt update” failed with EXPKEYSIG CD63F8B21266DE3C.

Diagnosis: The NVIDIA AI Workbench repository key expired on Dec 12, 2025, blocking the Wazuh installer from downloading dependencies.

The Action:

- Identified the expired key fingerprint: CD63F8B21266DE3C.

- Purged the legacy expired entry from the system keyring.

- Fetches the new key and stored it in the modern /usr/share/keyrings/ directory to follow Ubuntu 24.04 best practices.

Verification Command: `gpg --show-keys /usr/share/keyrings/ai-workbench-desktop-key.gpg.`

```
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$ gpg --show-keys /usr/share/keyrings/ai-workbench-desktop-key.gpg
pub  rsa3072 2024-12-12 [SC] [expires: 2027-12-13]
     52ECD6F6597CFBD86D5BF4DB3CD63F8B21266DE3C
uid  svc-workbench <svc-workbench@nvidia.com>
```

Issue B: The "Exit Status 127" Ghost

Symptom: Sub-process /usr/bin/dpkg returned an error code (1).

Diagnosis: Broken post-install scripts (.postinst) from a previous failed run were "poisoning" the package database.

Action: Performed "package surgery" by deleting all files in /var/lib/dpkg/info/wazuh-* and running dpkg --configure -a to force a clean state.

Evidence:

```
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$ ls -l /var/lib/dpkg/info/wazuh-*
-rw-r--r-- 1 root root 190 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.conf files
-rw-r--r-- 1 root root 7633129 Dec 20 16:11 /var/lib/dpkg/info/wazuh-dashboard.list
-rw-r--r-- 1 root root 9180280 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.md5sums
-rwxr-xr-x 1 root root 2247 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.postinst
-rwxr-xr-x 1 root root 1801 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.postrm
-rwxr-xr-x 1 root root 2103 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.preinst
-rwxr-xr-x 1 root root 2079 May 5 2023 /var/lib/dpkg/info/wazuh-dashboard.prerm
-rw-r--r-- 1 root root 1665 Apr 30 2025 /var/lib/dpkg/info/wazuh-indexer.conf files
-rw-r--r-- 1 root root 80966 Dec 20 16:09 /var/lib/dpkg/info/wazuh-indexer.list
-rw-r--r-- 1 root root 105109 Apr 30 2025 /var/lib/dpkg/info/wazuh-indexer.md5sums
-rwxr-xr-x 1 root root 1761 Apr 30 2025 /var/lib/dpkg/info/wazuh-indexer.postinst
-rwxr-xr-x 1 root root 1258 Apr 30 2025 /var/lib/dpkg/info/wazuh-indexer.preinst
-rwxr-xr-x 1 root root 1084 Apr 30 2025 /var/lib/dpkg/info/wazuh-indexer.prerm
-rw-r--r-- 1 root root 26 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.conf files
-rw-r--r-- 1 root root 2239426 Dec 20 16:10 /var/lib/dpkg/info/wazuh-manager.list
-rw-r--r-- 1 root root 2734604 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.md5sums
-rwxr-xr-x 1 root root 11131 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.postinst
-rwxr-xr-x 1 root root 2147 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.postrm
-rwxr-xr-x 1 root root 5669 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.preinst
-rwxr-xr-x 1 root root 2348 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.prerm
-rw-r--r-- 1 root root 490 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.shlibs
-rw-r--r-- 1 root root 60 Apr 30 2025 /var/lib/dpkg/info/wazuh-manager.triggers
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$
```

4. Verification of Success

The Service Layer (System Health)

```
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$ sudo systemctl status wazuh-indexer wazuh-manager wazuh-dashboard
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-20 16:11:50 MST; 43min ago
     Docs: https://documentation.wazuh.com
    Main PID: 2420167 (java)
      Tasks: 197 (limit: 153561)
     Memory: 1.6G (peak: 1.6G)
        CPU: 1min 1.795s
    CGroup: /system.slice/wazuh-indexer.service
            └─2420167 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10

Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.OpenSearch (file:/usr/share/wazuh-indexer-8.11.0/lib/opensearch.jar)
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.OpenSearch
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: System::setSecurityManager will be removed in a future release
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: Dec 20, 2025 11:11:46 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: COMPAT locale provider will be removed in a future release
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: A terminally deprecated method in java.lang.System has been called
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Security (file:/usr/share/wazuh-indexer-8.11.0/lib/opensearch.jar)
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.Security
Dec 20 16:11:46 spark-7a18 systemd-entrypoint[2420167]: WARNING: System::setSecurityManager will be removed in a future release
Dec 20 16:11:50 spark-7a18 systemd[1]: Started wazuh-indexer.service - wazuh-indexer.

● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-20 16:12:18 MST; 43min ago
     Tasks: 334 (limit: 153561)
    Memory: 5.0G (peak: 5.0G)
       CPU: 59.512s
    CGroup: /system.slice/wazuh-manager.service
            └─2422217 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              2422218 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              2422219 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              2422222 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              2422225 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              2422266 /var/ossec/bin/wazuh-authd
              2422282 /var/ossec/bin/wazuh-db
              2422312 /var/ossec/bin/wazuh-execd
              2422326 /var/ossec/bin/wazuh-analysisd
              2422532 /var/ossec/bin/wazuh-syscheckd
              2422549 /var/ossec/bin/wazuh-remoted
              2422611 /var/ossec/bin/wazuh-logcollector
              2422663 /var/ossec/bin/wazuh-monitord
              2422716 /var/ossec/bin/wazuh-modulesd

Dec 20 16:12:10 spark-7a18 env[2422153]: Started wazuh-analysisd...
```

The Network Layer (Port Listening)

The Ports:

443: The Dashboard (Web UI).

9200: The Indexer (Database).

55000: The Manager API.

1514: The Agent communication port (Incoming security logs)

```
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$ sudo ss -tulnp | grep -E '443|9200|55000|1514'
tcp  LISTEN  0      511             0.0.0.0:443      0.0.0.0:*      users:((("node",pid=2423983,fd=19))
tcp  LISTEN  0      128             0.0.0.0:1514    0.0.0.0:*      users:((("wazuh-remoted",pid=2422549,fd=4))
tcp  LISTEN  0      2048            0.0.0.0:55000   0.0.0.0:*      users:((("python3",pid=2422217,fd=42))
tcp  LISTEN  0      4096            [::ffff:127.0.0.1]:9200  *:*      users:((("java",pid=2420167,fd=650))
tcp  LISTEN  0      2048            [::]:55000      [::]:*      users:((("python3",pid=2422217,fd=44))
sunil77@spark-7a18:~/Desktop/projects/ai-soc-analyst/wazuh-docker/single-node$
```

The Application Layer (API Connectivity)

localhost/app/wz-home#/overview/?_g=(filters:(),refreshInterval:(pause:1t,value:0),tim

Sign in

Overview

a

AGENTS SUMMARY

This instance has no agents registered.
Please deploy agents to begin monitoring your endpoints.

Deploy new agent

LAST 24 HOURS ALERTS

Critical severity0Rule level 15 or higher

High severity0Rule level 12 to 14

Medium severity180Rule level 7 to 11

Low severity886Rule level 0 to 6

ENDPOINT SECURITY

Configuration Assessment

Scan your assets as part of a configuration assessment audit.

Malware Detection

Check indicators of compromise triggered by malware infections or cyberattacks.

File Integrity Monitoring

Alerts related to file changes, including permissions, content, ownership, and attributes.

THREAT INTELLIGENCE

Threat Hunting

Browse through your security alerts, identifying issues and threats in your environment.

Vulnerability Detection

Discover what applications in your environment are affected by well-known vulnerabilities.

MITRE ATT&CK

Explore security alerts mapped to adversary tactics and techniques for better threat understanding.

SECURITY OPERATIONS

PCI DSS

Global security standard for entities that process, store, or transmit payment cardholder data.

GDPR

General Data Protection Regulation (GDPR) sets guidelines for processing of personal data.

HIPAA

Health Insurance Portability and Accountability Act of 1996 (HIPAA) provides data privacy and security provisions for safeguarding medical information.

NIST 800-53

National Institute of Standards and Technology Special Publication 800-53 (NIST 800-53) sets guidelines for federal information systems.

TSC

CLOUD SECURITY

Docker

Monitor and collect the activity from Docker containers such as creation, running, starting, stopping or pausing events.

Amazon Web Services

Security events related to your Amazon AWS services, collected directly via AWS API.

Google Cloud

Security events related to your Google Cloud Platform services, collected directly via GCP API.

GitHub

Monitoring events from audit logs of your GitHub organizations.

Office 365

Security events related to